Accurate Segmentation of Pine Wilt Disease in UAV Remote Sensing Imagery: From Dataset Construction to Vision–Language Segmentation Framework With Coarse–Fine Dual-Label Supervision

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***Abstract*—Pine wilt disease (PWD) poses a serious threat to the global security of pine forest resources. In complex field scenarios using uncrewed aerial vehicles (UAVs) for remote sensing image segmentation, the early diseased and small lesion targets often manifest as weak features consisting of several pixels. Existing visual–language segmentation models primarily focus on extracting features of the disease itself and often fail to capture such subtle pixel-level information, leading to a relatively high missed detection rate. Therefore, a coarse–fine layered semantically coupled segmentation network (CF-SCSNet) is proposed in this study, in which a core coarse–fine dual-label (CFL) loss strategy is incorporated as the key supervision module. Fine labels are used to precisely delineate the actual lesion areas and guide the model in learning detailed texture features. Coarse labels are generated by extending based on the fine labels and are used to cover both the lesion and its immediate surroundings (not the entire image), enabling the model to learn spatial correlation patterns between the lesion and the adjacent environment. Furthermore, to accurately capture the spatial distribution patterns of the disease, a pixel-aware context aggregation (PACA) module is designed based on text priors, which enhances local spatial relationship learning. Simultaneously, a high semantic fusion (HighSF) module is proposed. Through complementary differential receptive field designs, it effectively captures disease features at various infection phases. Finally, a PWD remote sensing image dataset is constructed, covers multiple geographical regions and is adaptable to complex field imaging scenarios. Extensive experimental results demonstrate that CF-SCSNet achieves outstanding performance with an mIoU of 83.55% on the self-built PWD dataset. To evaluate its generalization capability, further validation is conducted by using two public remote sensing segmentation datasets (FarmSeg and RRSIS), confirming the effectiveness and robustness of CF-SCSNet across different scenarios and tasks. The code is available at https://github.com/TechJots-Liu/CF-SCSNet**

***Index Terms*—Coarse–fine dual-label (CFL), multimodal fusion (MMF), pine wilt disease (PWD), semantic segmentation, visual–language.**

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I. Introduction

THE ecological integrity of forest ecosystems, in their capacity as global carbon sinks and biodiversity refuges, is inextricably linked to the foundations of sustainable human development [1], [2]. However, global forest health is under increasing threat, among which pine wilt disease (PWD) has become one of the world's most dangerous forest diseases due to its severe destructiveness, rapid transmission, and widespread [3]. The dynamics of forest pests and diseases are not only an important bio-indicator of ecosystem imbalance, but also a central challenge for sustainable forest management and early warning of biological disasters in the context of global change.

Conventional forest disease detection techniques, primarily relying on manual ground inspections and fixed sample plot surveys, are limited in scope and timeliness, hindering comprehensive and efficient surveillance. In contrast, the advent of remote sensing technology, with its multiplatform and multiscale observation capabilities, has revolutionized forest health detection. In particular, the integration of deep learning architectures (such as convolutional neural networks (CNNs) and Transformers) has significantly advanced remote sensing image segmentation tasks [4], including those focused on PWD detection [5], [6], [7].

Despite these advancements, critical challenges persist. A significant bottleneck for detection is posed by complex understory environmental interference [8], [9], coupled with the small-target characteristics of early-stage and latent-stage infected trees (with canopy widths of only 1–5 m) [10]. Although hyperspectral and multispectral techniques have improved identification accuracy in certain contexts [11], [12], [13], their effectiveness remains limited for early-stage infections due to spectral heterogeneity and high equipment costs. Moreover, the subpixel characteristics of early-stage diseased trees often lead to geometric mismatches and feature separability degradation [14].

Recent paradigm shifts in computer vision have prompted the integration of self-attention mechanisms and Transformer architectures into CNNs, enabling better capture of long-range

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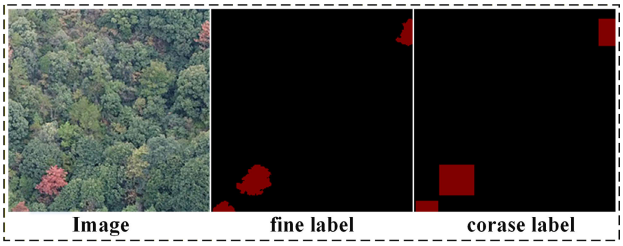


Fig. 1. Example of fine label and the corresponding coarse label.

dependencies and global context [15], [16]. Early Transformer-based PWD detection models primarily utilized unimodal data, but growing interest has turned to multimodal fusion (MMF) frameworks that combine, for example, uncrewed aerial vehicle (UAV) RGB and hyperspectral data [17]. However, issues such as insufficient cross-modal alignment and high computational complexity continue to impede practical application.

Inspired by the success of vision–language models (VLMs) in natural image understanding, researchers have begun exploring language-guided segmentation in remote sensing [18], [19]. Early efforts focused on multimodal feature fusion mechanisms [20], [21], [22], while later studies introduced more efficient cross-modal architectures such as the vision-and-language Transformer (ViLT) [23]. Despite these advances, existing visual–language segmentation models primarily focus on extracting features of the disease itself and often fail to capture such subtle pixel-level information, leading to a relatively high missed detection rate.

To address the above limitations, this study proposes a coarse–fine layered semantically coupled segmentation network (CF-SCSNet). Specifically, the network introduces a coarse–fine dual-label (CFL) loss strategy, thereby establishing a dual-oriented approach for precise positioning and association learning. Fine labels are used to precisely delineate the actual lesion areas and guide the model in learning detailed texture features. Coarse labels are generated by extending based on the fine labels and are used to cover both the lesion and its immediate surroundings (not the entire image), as shown in Fig. 1, enabling the model to learn spatial correlation patterns between the lesion and the adjacent environment, thereby reducing the false-negative rates of early-stage lesions and small target lesions. Furthermore, to accurately capture the unique spatial distribution pattern of PWD, we designed a pixel-aware context aggregation (PACA) module based on textual priors. This module relies on a vertical–horizontal bidirectional feature decomposition strategy, which disentangles the intertwined spatial information from 2-D features into two independent 1-D structured features. It can therefore precisely capture both the vertical gradient of the disease spreading downward from the treetop and its irregular patchy distribution in the horizontal direction, enhancing local spatial correlation learning while reducing computational load. At the same time, we specifically propose a high semantic fusion (HighSF) module, which employs dilated convolutions with different dilation rates to construct a complementary mechanism for differentiated receptive fields. This design enables comprehensive coverage of all-scale disease features, effectively solving the technical bottlenecks of missed detection of small lesions and blurred boundary localization of large diseased areas.

The main contributions of this work are summarized as:

1) To break through unimodal limitations and enable high-level image–text fusion for fine-grained remote sensing interpretation, this article presents a novel dataset coupling UAV remote sensing images with expert-guided textual descriptions. Unlike generic image–text pairs, these texts are semantically anchored to diseased regions, directly supporting cross-modal alignment of fine-grained spatial features (critical for small target segmentation) and laying the foundation for coarse–fine semantic coupling.
2) To tackle the dual challenges of fine-scaled, sparsely distributed diseased regions and cluttered backgrounds, we propose a CF-SCSNet. The network is driven by a CFL loss strategy and works in tandem with the custom PACA and HighSF modules, achieving precise lesion localization and learning spatial correlations around lesions. This integration is designed to reduce the missed detection rate of early-stage lesions and small targets.
3) To verify the generalization performance of CF-SCSNet, we evaluated the model on the farmland segmentation (FarmSeg) dataset and the urban building segmentation dataset (RRSIS). Results demonstrate that CF-SCSNet achieves significantly superior performance compared to existing segmentation methods.

II. Related Work

A. Remote Sensing-Based Forest Disease Detection

Traditional forest disease detection has largely depended on manual surveys, which are labor-intensive and spatially limited. The proliferation of remote sensing technology has enabled large-scale, periodic forest health assessment. Early approaches utilized vegetation indices derived from multispectral and hyperspectral data to detect biochemical changes in infected trees. For example, red-edge indices such as NDRE and RENDVI have achieved over 90% accuracy in detecting mid-to-late-stage infections, though their performance drops significantly for early-stage infections (accuracy less than 77%) [12], [13]. Multispectral techniques that integrate vegetation indices with dynamic convolution have improved detection performance, reaching an mAP of 89.1% [11], [25]. However, these methods remain limited by spectral heterogeneity in complex forest stands [26].

With the rise of deep learning, CNNs and Transformers have been widely applied to PWD segmentation tasks [5], [6], [7]. These models excel at extracting multilevel spatial and spectral features, yet they struggle with small targets and complex backgrounds. To mitigate these issues, strategies such as center-circle sampling [14], multiscale feature enhancement [27], [28] have been proposed, reducing missed detection rates to 6.6% in some cases [29]. Nonetheless, the accuracy for latent-stage infections (needle discoloration less than 10%) generally remains below 70%, underscoring the limitations of relying solely on local visual features. Dual-time phase analysis using digital surface models (DSMs) has also been explored to monitor logging activities and forest changes, achieving 77.91% accuracy in elevation-based detection [30].

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However, such methods require multitemporal data, increasing operational costs and complexity.

B. VLMs in Remote Sensing

VLMs offer a transformative approach to remote sensing interpretation by integrating the intuitive perception of visual data with the semantic richness of linguistic descriptions. This synergy enables models to move beyond pixel-level analysis toward a more conceptual understanding of scene content, which is particularly valuable for complex tasks such as forest disease detection and segmentation, where contextual and semantic cues are critical.

The development of large-scale VLMs, exemplified by CLIP [31], has accelerated their adoption in remote sensing tasks. Initial efforts focused on adapting these frameworks to geospatial contexts. For instance, ChangeCLIP [15] demonstrated the feasibility of leveraging linguistic guidance for targeted change detection in multitemporal imagery. Subsequent research explored text-guided segmentation through various MMF mechanisms [20], [21], [22], though these early approaches often struggled to achieve precise cross-modal alignment while maintaining computational efficiency.

To address the computational constraints of early VLM architectures, the ViLT [23] introduced a more efficient design using linear projections instead of convolutional encoders. However, this simplification proved inadequate for capturing subtle spectral-spatial variations characteristic of forest environments. Recent developments have therefore shifted toward hybrid architectures [32], [33], [34] that combine convolutional and attention mechanisms, along with contrastive learning strategies [35], [36] to enhance cross-modal representation learning. These advancements have improved the model's ability to handle modality-specific characteristics while preserving local feature details.

A parallel line of research has investigated the integration of domain knowledge through structured textual descriptions. Several studies [37], [38], [39], [40] have incorporated rich semantic priors to improve segmentation accuracy, particularly for challenging targets like early-stage infected trees. Nevertheless, these methods typically depend on meticulously annotated text descriptions with precise spatial and attribute information, which are difficult to scale for practical applications. To bridge the semantic gap between visual and linguistic modalities, recent approaches [41], [42] have proposed decoupling object concepts from spatial contexts, enabling more precise pixel-level alignment through sophisticated decoding mechanisms.

Despite these advancements, applying VLMs to forestry disease detection and segmentation still faces two core challenges.

1) In forestry remote sensing images, early disease symptoms and small target plants typically constitute a very small pixel proportion, while the complex backgrounds inherent to wild scenes (e.g., healthy vegetation and terrain shadows) present an abundance of high-dimensional distracting features. Most existing VLMs primarily focus on generic feature fusion strategies and lack specific optimization for the peculiarities of forestry remote sensing scenarios.
2) In forestry scenarios, the occurrence and spread of diseases are significantly spatially correlated with surrounding vegetation growth, terrain, and the distribution of associated plants. This correlation is an important basis for locating weak feature lesions and accurately segmenting disease areas. However, most current studies only focus on extracting visual features of the disease itself, ignoring this core correlation.

III. Materials

A. Data Introduction

To address the limitations of current datasets for PWD segmentation, which are often restricted by single geographical locations and fixed imaging conditions, this study constructed a PWD remote sensing image dataset that covers multiple geographical regions and is adaptable to complex field imaging scenarios, supporting the training and validation of highly generalizable segmentation models. The dataset comprises 1803 visible-light images (224 $\times$ 224 pixels) captured by UAVs. Among these, 30% of the images were acquired in Yunnan Province, China (Malipo County and Wenshan City; 103°43′–105°18′ E, 22°48′–23°44′ N). The remaining 70% were developed by Xu et al. [11], captured with a DJI M300 in the pine forests surrounding Dajiaofang Reservoir, Dongzhou District, Fushun City, Liaoning Province, China (124°1′–124°24′ E, 41°16′–42°0′ N), as shown in Fig. 2.

B. Data Processing

Data preprocessing involved three main steps: image screening, size normalization, and data expansion via cropping. First, all collected images underwent uniform screening to discard invalid samples that were blurry, overexposed, or contained ambiguous subjects. To prevent overfitting and accelerate training, a sliding-window cropping strategy was then applied. A fixed 224 $\times$ 224-pixel window traversed each image with a stride of 100 pixels, ensuring 30% overlap between adjacent subimages. This overlap facilitated comprehensive local feature capture while increasing sample size. Following cropping, manual visual inspection removed subimages either lacking signs of PWD or exhibiting forest coverage below 20%. Finally, the resulting set of 1803 high-quality images was randomly divided into training, validation, and test subsets in a 7:2:1 ratio for subsequent model training and evaluation.

C. Data Annotation

1) Image Annotation: First, pixel-level annotation of the images was performed using the LabelMe software to obtain the fine labels. Suspected disease areas of uncertain status were uniformly labeled under a "pending verification" category and were then collectively reviewed upon completion of the annotation round to finalize their classification, thereby improving consistency and reducing subjective error. Subsequently, to ensure data accuracy and usability, a cross-validation mechanism was implemented in which different annotators reviewed each other's work. This process focused on identifying and correcting incorrect or missed annotations, as well as boundary deviations. Through this stringent screening, a total of

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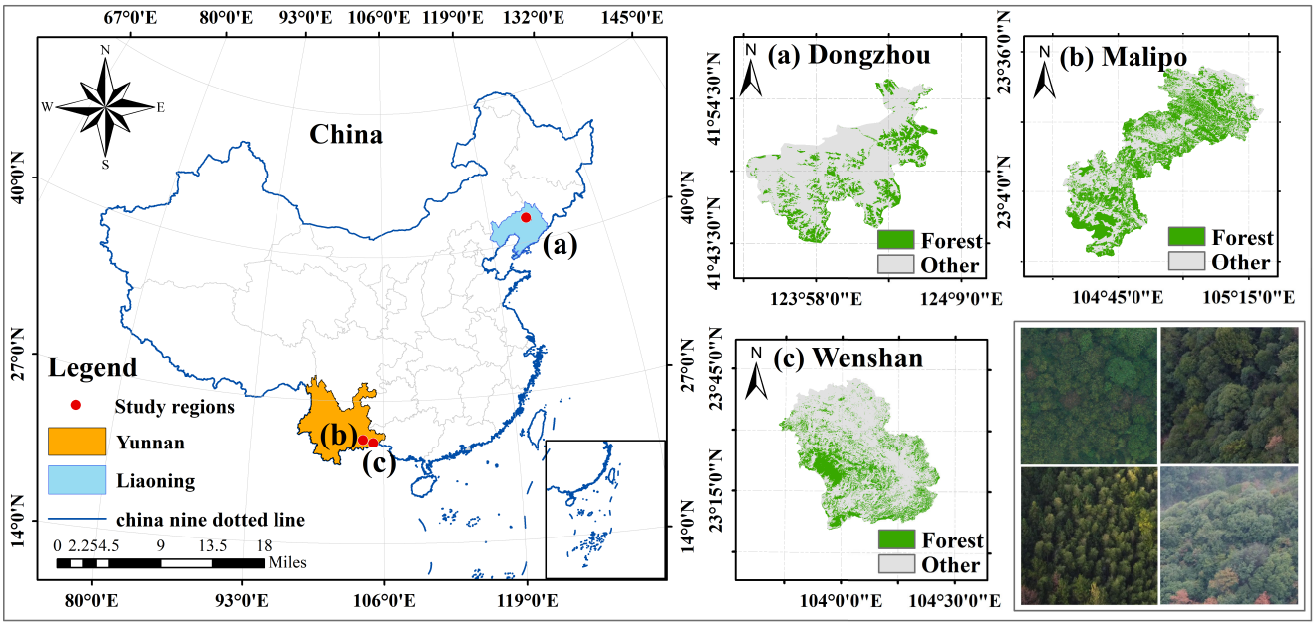


Fig. 2. Overview of the study areas in the PWD dataset, highlighting the two sampled provinces, three study areas, and representative image examples.

1803 valid images with clear texture and dominant disease features were retained. Furthermore, to adapt to the coarse and fine label collaborative training framework (CF-SCSNet) developed in this study, based on the irregular fine label areas annotated manually, code is generated through bounding boxes to extract the enclosing rectangles to generate regular coarse labels, as illustrated in Fig. 1.

2) Text Annotation: For text annotation, we produced structured textual descriptions of the affected areas based on their location and relative size (measured in pixel percentage) using an in-house developed text-generation code. Specifically, the text-generation algorithm takes mask labels as input and populates predefined sentence templates by computing the number of instances, along with their orientation and spatial aggregation, thereby generating consistent descriptions in batches, as illustrated in Fig. 3. To ensure the consistency of the automated method in both format and content, we randomly selected 10% of the samples (totaling 180 image–text pairs) for independent blind review by two team members. The evaluation criterion was whether directional and size terms in the text accurately and unambiguously described the affected area. In cases of ambiguous descriptions, a third expert made the final determination. Three rounds of random sampling showed that the average agreement rate between the two reviewers exceeded 95%, directly confirming the high reliability of the automated annotation process.

IV. Methodology

To address the lack of textual a priori knowledge of existing VLMs for remote sensing images of PWD. In this study, a PWD dataset was constructed to solve this problem. In addition, CF-SCSNet is designed by incorporating textual modalities into a single image segmentation model. In this section, the basic principles and processes of constructing a PWD dataset and CF-SCSNet are studied in depth.

A. VLM for the Segmentation of PWD

1) Overall Framework: To address the multisource information fusion requirements in UAV remote sensing segmentation of forest pests and diseases, a cross-modal CF-SCSNet is proposed in this study, as illustrated in Fig. 4. The model architecture follows an encoder–decoder framework. First, forestry remote sensing images captured by UAV and pathological text descriptions are fed in as inputs. Image features (V_1, V_2, V_3, V_4) are extracted via a CNN image encoder. Simultaneously, text features (T) are extracted using a RoBERTa [24] text encoder. Subsequently, these image and text features are deeply fused to obtain the integrated features y_1, y_2, y_3, y_4. In the decoder stage, the PACA and HighSF modules are embedded into the processing flow to learn the spatial distribution patterns of disease features and to capture their sizes at different scales. Finally, through upsampling and skip connections, the spatial resolution of the feature maps is progressively restored to produce the fine-grained prediction $(pred_f)$. To fulfill the model's dual-output requirement, feature y_4 is also upsampled to the output dimension, yielding the coarse-grained prediction $(pred_c)$.

2) Image–Text Fusion Encoding: The model employs an MMF strategy [33] during the encoding stage. Given that shallow features retain closer proximity to raw perceptual data, MMF adopts a multihead cross-attention mechanism in the first encoding stage, where image features V_1 serve as the query and text features T act as the key and value, guiding the focus toward text-relevant image regions to achieve deep interactive fusion. It is noteworthy that this direct cross-modal attention is applied only between text and the shallow features V_1. In subsequent encoding layers (2, 3, and 4), the output y_{i-1} from the previous layer is concatenated with the corresponding-scale image features V_i, without further text interaction. The fusion process at the first layer is formalized

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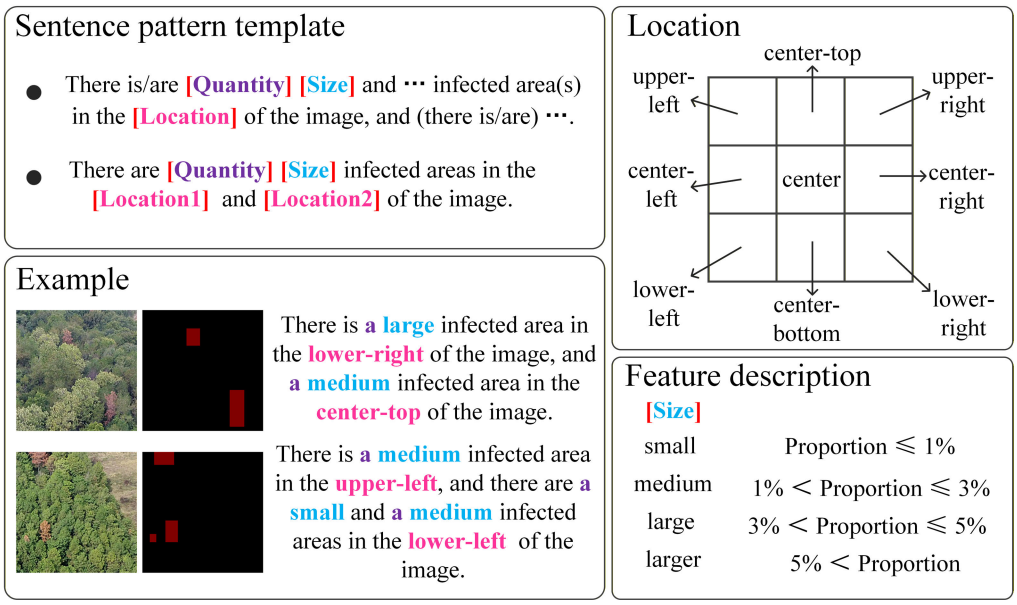


Fig. 3. Framework for generating structured text descriptions of disease areas in remote sensing images, encompassing sentence templates, image region division rules, definitions of disease area size characteristics, and example outputs.

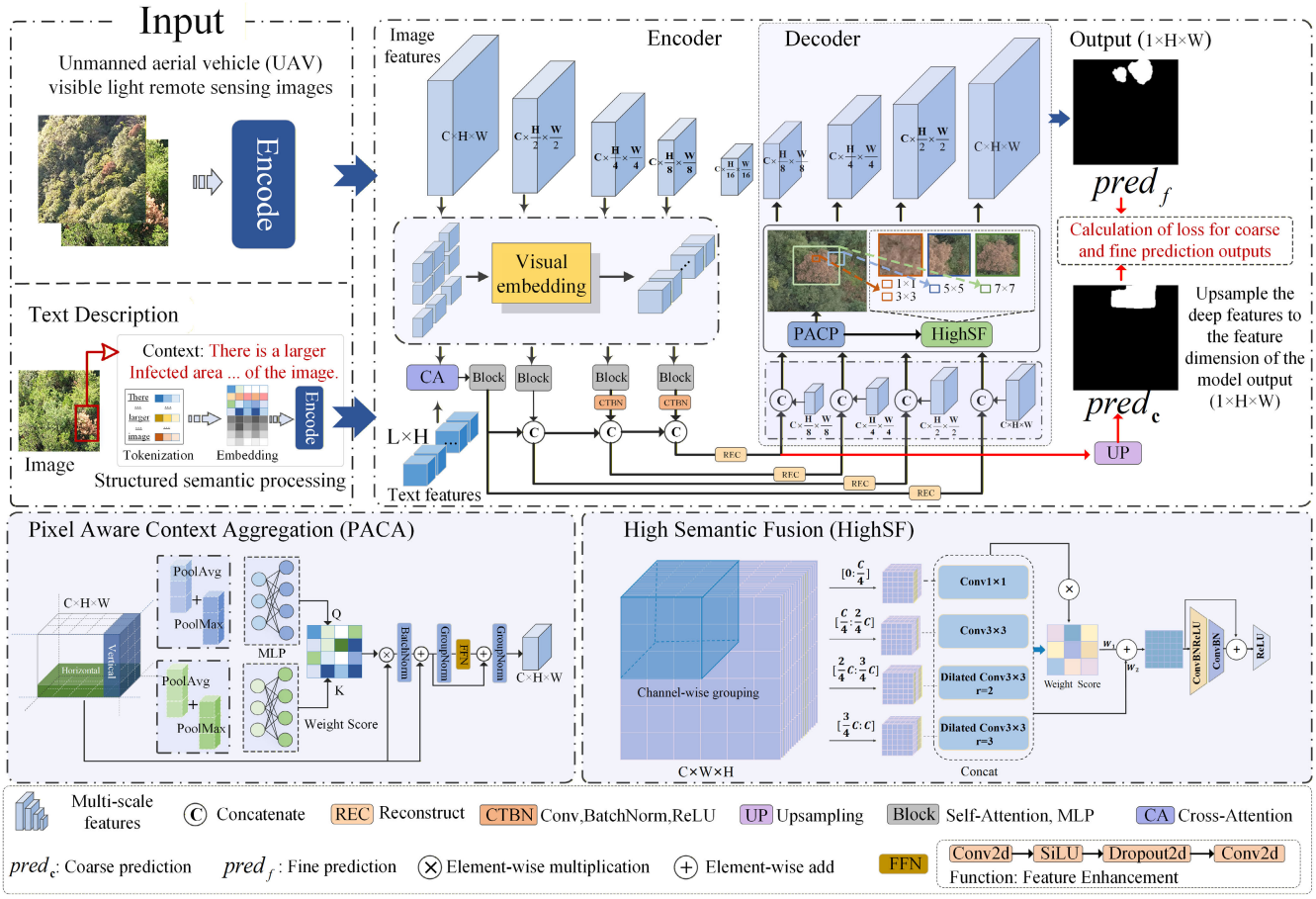


Fig. 4. Overview of the proposed framework. The model takes bimodal inputs and produces two final outputs of $pred_c$ and $pred_f$

as follows:

$$V_1' = \text{Embeddings}\,(V_1) \tag{1}$$

$$y_1 = \text{Block}\left(\text{MMF}\left(V_1', T\right)\right) \tag{2}$$

where V_1 represents the image features, T represents the Roberta-base processed text features, and V_1' represents the embedded features of V_1. The embedding module is employed to transform 2-D images into 1-D feature sequences that contain spatial structure information. These feature sequences subsequently provide inputable feature forms for attention mechanisms. The block module consists of a self-attention mechanism and a feed-forward neural network multilayer per-

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ceptron (MLP), designed to iteratively optimize the sequence features after image embedding, thereby capturing remote dependencies in the sequence and simultaneously mitigating gradient vanishing. Each subsequent fusion layer receives the output features of the preceding layer and image features at the corresponding scale. The process is formalized as follows:

$$V_i' = \text{Embeddings}\left(V_i\right) \tag{3}$$

$$y_i = \text{Concat}\left(\text{CBR1d}\left(\text{Block}\left(V_i'\right), y_{i-1}\right)\right) \tag{4}$$

where $i \in \{2,3,4\}$. The CBR1d block (composed of Conv1d, BatchNorm1d, and ReLU) projects the image and text features into a shared dimensional space.

Then, the feature y_i ($i \in \{1,2,3,4\}$) is first reconstructed to match the spatial dimensions of the corresponding image feature V_i, followed by channel-wise concatenation to yield $V_{i,\text{c}}$. This helps to avoid model performance fluctuations caused by inaccurate text annotations. Notably, the reconstructed feature y_4' is upsampled to match the channel count of the model's prediction output, generating the rough prediction features $pred_{\text{c}}$ (as detailed in Fig. 4). The process is formalized as follows:

$$y_i' = \text{Reconstruct}\left(\text{S}_{\text{HW}}\left(y_i\right)\right) \tag{5}$$

$$V_{i,c} = \text{Concat}\left(y_i', V_i\right) \tag{6}$$

$$\text{S}_{\text{HW}}\left(y\right) = y \rightarrow y_{\text{reshaped}}\left(B, C, \sqrt{N}, \sqrt{N}\right) \tag{7}$$

$$pred_{\text{c}} = \text{Up}\left(y_4'\right) \tag{8}$$

where $y_i \in \mathbb{R}^{B\times N\times C}, i \in \{1,2,3,4\}$. B, C, and N represent the batch size, the number of channels, and the size of the embedded patch, respectively. The reconstruct module converts sequence-form features into image dimensions, consisting of a sequence-to-space mapping function $S_{\text{HW}}(y)$, upsampling, convolution (Conv), BatchNorm (BN), and activation (ReLU). The Up represents the upsampling operation.

3) PACA Module: Although the fusion features enhance cross-modal semantic associations, the spatial signals from healthy and lesion areas within the 2-D fusion features interfere with each other, making it difficult to precisely separate the characteristic patterns of vertical spread and horizontal sheet-like distribution of the disease. Moreover, the text serves only as global background information and cannot provide pixel-level guidance.

To address this challenge, the PACA module decouples the spatial information intertwined within the 2-D fusion features into independent vertical and horizontal 1-D structured features (as illustrated in Fig. 4). This decoupling strategy can precisely capture the vertical lesion gradient and horizontal distribution range, thus mitigating the mutual interference between healthy and lesion tissue information in the original 2-D features. At the same time, decomposing the 2-D features into two 1-D structured features can effectively reduce the computational load.

First, the input feature tensor V is decomposed along the spatial dimension into vertical-wise feature subspace V_{h} and horizontal-wise feature subspace V_{w}, enabling multidirectional feature decoupling. To capture the global feature distribution of diseased trees under varying lighting and viewing angles, we employ BN for cross-scenario feature alignment, combined with GroupNorm (GN) to enhance the capture of local lesion details, thereby forming a complementary normalization strategy. The specific formulation is as follows:

$$V \in \mathbb{R}^{B\times C\times\Omega} \xrightarrow[\text{axis decomposition}]{\text{spatial}} V_{\text{h}}, V_{\text{w}} \tag{9}$$

$$V_{\text{h}} = \left(p_{\text{avg}}^{\text{h}} + p_{\text{max}}^{\text{h}}\right) : \mathbb{R}^{B\times C\times\Omega} \rightarrow \mathbb{R}^{B\times C\times\Omega_{\text{h}}} \tag{10}$$

$$V_{\text{w}} = \left(p_{\text{avg}}^{\text{w}} + p_{\text{max}}^{\text{w}}\right) : \mathbb{R}^{B\times C\times\Omega} \rightarrow \mathbb{R}^{B\times C\times\Omega_{\text{w}}} \tag{11}$$

where $\Omega = \text{H} \times \text{W}$ represents the spatial domain, $\Omega_{\text{h}} = \text{H} \times 1$ and $\Omega_{\text{w}} = 1 \times \text{W}$ represent the vertical and horizontal subdomains, respectively, and $\oplus$ represents tensor addition. $p_{\text{avg}}^{\text{h}}, p_{\text{max}}^{\text{h}}, p_{\text{avg}}^{\text{w}}, p_{\text{max}}^{\text{w}}$ represent the average and max-pooling operators along the vertical and horizontal directions, respectively.

These features are then projected into a high-dimensional space using learnable weight matrices, and their interactions are realized via bilinear products to establish long-distance dependencies between pixels. A spatial attention mask matching the input feature dimensions is generated through ReLU activation, enabling adaptive feature enhancement. The specific formulation is as follows:

$$A = \text{RELU}\left(\text{MLP}_{\text{h}}\left(V\right) \cdot \text{MLP}_{\text{w}}\left(V\right)\right) \tag{12}$$

$$V' = A \odot V \tag{13}$$

where V' represents the augmented features. MLP_{h} and MLP_{w} are MLPs for vertical and horizontal features, each formed by a sequential stack of LayerNorm (LN), followed by a SiLU activation, and a linear layer ($\text{Linear}_{c\rightarrow c}$). $\odot$ for element-wise multiplication.

To effectively balance global feature alignment across scenes with the discriminative needs of local lesion features, and to address statistical bias in small-batch training, the enhanced feature V' undergoes dual projection via BN and GN for global and local processing, respectively, yielding the refined feature V''. V'' is then fed into a feed-forward network (FFN) module, which performs nonlinear transformations and multiscale feature fusion to enhance semantic representation. The specific formulations are as follows:

$$V'' = \text{GN}\left(V \oplus \text{BN}\left(V'\right)\right) \tag{14}$$

$$\widehat{V} = \text{GN}\left(V'' \oplus \text{FFN}\left(V''\right)\right) \tag{15}$$

$$\text{BN}\left(x_{i,j,c}\right) = \gamma_c \cdot \frac{x_{i,j,c} - \mu_{\text{B},c}}{\sqrt{\sigma_{\text{B},c}^2 + \epsilon}} + \beta_c \tag{16}$$

$$\text{GN}\left(x_{i,j,c}\right) = \gamma_c \cdot \frac{x_{i,j,c} - \mu_{\text{g}(c)}}{\sqrt{\sigma_{\text{g}(c)}^2 + \epsilon}} + \beta_c \tag{17}$$

$$\mu_{\text{B},c} = \frac{1}{N \cdot H \cdot W} \sum_{n=1}^{N} \sum_{i=1}^{H} \sum_{j=1}^{W} x_{n,i,j,c} \tag{18}$$

$$\mu_{\text{g}(c)} = \frac{1}{H \cdot W \cdot (C/\text{G})} \sum_{k\in\text{g}(c)} \sum_{i=1}^{H} \sum_{j=1}^{W} x_{i,j,k} \tag{19}$$

where $x_{i,j,c}$ represents the value of channel C at position (i, j) in the input feature map, γ_c and β_c are learnable parameters, and $\sigma_{\text{B},c}^2$ and $\sigma_{\text{g}(c)}^2$ represent the variance statistics computed over the batch and within groups, respectively. Similarly, $\mu_{\text{B},c}$ and $\mu_{\text{g}(c)}$ represent the batch-wise and group-wise mean statistics, respectively. Specifically, BN computes the mean and

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variance across the entire batch during training, whereas GN partitions the channels into G groups and computes statistics within each group, where G represents the number of groups.

4) HighSF Module: During the decoder's feature sampling, multiscale features from different encoder levels are typically integrated. Direct fusion operations (e.g., concatenation or addition) often lead to the loss of fine details, especially for deeper-level features. Therefore, except for the shallowest features, which adopt concatenation, we employ the HighSF module to fuse the remaining three (deeper-level) feature layers. This approach enables finer integration of multiscale representations, effectively mitigates detail loss, and maintains computational efficiency.

For the decoder of the proposed HighSF module, simple concatenation fails to fully leverage the complementary information from multilayer features in encoder skip connections. Thus, we adopt a hierarchical feature fusion strategy (as illustrated in Fig. 4). The HighSF module is employed to address the limitation of single-path receptive fields. It enables fine-grained fusion through multiscale dilated convolution, spatial attention mechanisms, and dynamic weight aggregation. At the end of the decoder, a concatenation operation is adopted to balance computational efficiency and semantic integrity, thereby facilitating the rapid recovery of spatial details.

Algorithm 1 Workflow of PACA and HighSF

Input: Updated feature state : $V_{\text{up}} \in \mathbb{R}^{B\times C\times H/2\times W/2}$, The feature state : $V_{i,\text{c}} \in \mathbb{R}^{B\times C\times H\times W}$;
Output: Updated feature state : $V_{\text{up}} \in \mathbb{R}^{B\times C\times H\times W}$;
$V'_{\text{up}} \leftarrow \text{Upsample}(V_{\text{up}}) \in \mathbb{R}^{B\times C\times H\times W}$;
/* Use PACA to enhance the current features */
$V'_{i,\text{c}} \leftarrow \text{PACA}(V_{i,\text{c}}) \in \mathbb{R}^{B\times C\times H\times W}$;
/* The HighSF algorithm is used at the end of the decoder */
if C != 64 **then**
    $V_{\text{SF}} \leftarrow \text{HighSF}(V'_{\text{up}}, V'_{i,\text{c}}) \in \mathbb{R}^{B\times C\times H\times W}$;
    $V'_{\text{SF}} \leftarrow \text{CBR2d}(\text{in}_\text{c} = C, \text{out}_\text{c} = C/2)(V_{\text{SF}}) \in \mathbb{R}^{B\times C/2\times H\times W}$;
    $V''_{\text{SF}} \leftarrow \text{CBR2d}(\text{in}_\text{c} = C/2, \text{out}_\text{c} = C/2)(V'_{\text{SF}}) \in \mathbb{R}^{B\times C/2\times H\times W}$;
    Append V''_{SF} to V_{up};
    return: V_{up};
else
    $V_{\text{CF}} \leftarrow \text{Concat}([V'_{\text{up}}, V'_{i,\text{c}}], \text{dim} = 1) \in \mathbb{R}^{B\times 2C\times H\times W}$;
    $V'_{\text{CF}} \leftarrow \text{CBR2d}(\text{in}_\text{c} = 2C, \text{out}_\text{c} = C)(V_{\text{CF}}) \in \mathbb{R}^{B\times C\times H\times W}$;
    $V''_{\text{CF}} \leftarrow \text{CBR2d}(\text{in}_\text{c} = C, \text{out}_\text{c} = C)(V'_{\text{CF}}) \in \mathbb{R}^{B\times C\times H\times W}$;
    Append V''_{CF} to V_{up};
    return: V_{up};
end if= 0

To help readers better understand the interaction logic between the HighSF and PACA, we provide the detailed pseudocode in Algorithm 1.

First, the upsampled feature from the previous decoder layer V'_{up} and the skip connection features $V'_{i,c}$ from the corresponding encoder stage i are concatenated along the channel dimension to construct an enriched feature representation. The concatenated features are then processed through a 1×1 convolution layer for cross-channel information integration and dimension reduction, producing a compact feature representation V_{cat}.

V_{cat} is partitioned into four equal parts along the channel dimension, creating four feature subspaces V^j where $j \in 1, 2, 3, 4$. Each subspace V^j is processed in parallel using convolutional operations with different kernel sizes and varying dilation rates to capture multiscale contextual information across different receptive fields. Outputs from these parallel convolutional streams are concatenated along the channel dimension, forming the final multiscale feature representation V'_{cat} that integrates both local details and global context. The details are shown in the following equations:

$$V_{\text{cat}} = \text{ConvBN}\left(Concat\left(V'_{\text{up}}, V'_{i,c}\right)\right) \tag{20}$$

$$V^j = V_{\text{cat}}\left[:, \frac{C\times(j-1)}{4} : \frac{C\times j}{4}, :, :\right] \tag{21}$$

$$\text{ConvF}\left(V^j\right) = \begin{cases} Kernel_size = 1, & j = 1, d = 1 \\ Kernel_size = 3, & j = 2, d = 1 \\ Kernel_size = 3, & j = 3, d = 2 \\ Kernel_size = 3, & j = 4, d = 3 \end{cases} \tag{22}$$

$$V^j_{\text{cat}} = \sigma\left(\text{BN}\left(\text{ConvF}\left(V^j\right)\right)\right) \tag{23}$$

$$V'_{\text{cat}} = \text{Concat}\left(V^1_{\text{cat}}, V^2_{\text{cat}}, V^3_{\text{cat}}, V^4_{\text{cat}}\right) \tag{24}$$

where V'_{up} represents the upsampled output features from the decoder layer, $V'_{i,c}$ represents the skip connection from the encoder, and C represents the number of channels (C must satisfy $C \mod 4 = 0$). ConvF(V^j) represents the convolution kernel defined for the four subspaces and d is the dilation rate of convolution. BN is batch normalization, σ is the SiLU activation function, and V^j_{cat} represents the multiscale features obtained through convolution operations with different kernels.

To learn channel-wise importance weights from the feature map, a channel attention mechanism is designed to emphasize critical channels while suppressing irrelevant ones. Specifically, global average pooling (GAP) is first applied to V'_{cat} to extract global statistical features for each channel, generating channel attention weights σ_{weights}. The original features V'_{cat} are then weighted element-wise via the Hadamard product. This process is formulated as

$$V''_{\text{cat}} = \mathbf{F} \odot \sigma\left(\Gamma\left(\frac{1}{H\times W}\sum_{i=1}^{H}\sum_{j=1}^{W}\mathbf{F}_{:,:,i,j}\right)\right) \tag{25}$$

where $\mathbf{F}$ represents V'_{cat}, $\Gamma(\cdot)$ represents the 1-D convolution transform, and V''_{cat} is the channel-weighted enhanced feature. The enhanced feature V''_{cat} and V'_{cat} are then combined via a dynamically weighted summation controlled by a learnable parameter ρ, enabling complementary integration of the two feature representations. This fusion operation is defined as

$$V'''_{\text{cat}} = V'_{\text{cat}} \cdot (1-\rho) + V''_{\text{cat}} \cdot \rho \tag{26}$$

where ρ is the dynamic weight factor. Finally, the fused features undergo a nonlinear transformation and optimization via a residual block structure. The calculation of V_{out} is defined as

$$V_{\text{out}} = \text{ReLU}\left(\text{BN2d}\left(\text{Conv2d}\left(\text{CBR1d}\left(V'''_{\text{cat}}\right)\right)\right) + V'''_{\text{cat}}\right) \tag{27}$$

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where V_{out} is the final output feature and BN2d is the abbreviation for the BatchNorm2d.

5) CFL Loss Function: The framework consists of two supervisory branches: a coarse-grained branch and a fine-grained branch (as illustrated in Fig. 5). In the coarse-grained branch, the model first upsamples the deep feature y_4 from the fusion module to generate the coarse-grained prediction $Label_{pred,c}$. The loss term L_c then calculates the difference between this prediction and the coarse-grained ground truth $Label_{truth,c}$. This provides robust constraints at the local semantic level, guiding the model to anchor the lesion and its surrounding context while avoiding ineffective learning from the global background. In the fine-grained branch, the decoder produces the fine-grained prediction $Label_{pred,f}$ via progressive upsampling as the model's final output. The loss term L_f calculates its difference against the fine-grained ground truth $Label_{truth,f}$, thereby constraining the model to precisely fit the pixel-level textures and boundaries of lesions, which enhances segmentation accuracy. The total loss is obtained by a weighted combination of the two loss terms, formulated as

$$\mathcal{L}_{\text{BCE}} = \frac{\alpha}{|P|+\sigma} \sum_{\substack{i \\ y_i>0.5}} \text{BCE} + \frac{(1-\alpha)}{|N|+\sigma} \sum_{\substack{i \\ y_i\le 0.5}} \text{BCE} \tag{28}$$

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\cdot(w\odot\hat{y})\cdot(w\odot y)+\sigma}{(w\odot\hat{y})\cdot(w\odot\hat{y})+(w\odot y)\cdot(w\odot y)+\sigma} \tag{29}$$

$$\mathcal{L}_{\text{Total}} = \beta\cdot\mathcal{L}_{\text{BCE}} + (1-\beta)\cdot\mathcal{L}_{\text{Dice}} \tag{30}$$

where $\hat{y}$ and y denote the prediction and truth labels in vector form, respectively. The weight parameters α and β, which control the proportion of positive sample loss, are set to 0.5 by default. The positive sample index set is defined as $P = \{i \mid y_i > 0.5\}$ and the negative set is defined as $N = \{i \mid y_i < 0.5\}$, where $|P|$ and $|N|$ indicate their cardinalities. The standard BCE loss is given by $\text{BCE}(\hat{y}_i, y_i) = -[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)]$, $\hat{y}_i$ and y_i represent the predicted probability (scalar) and ground-truth label (0 or 1) of the ith pixel after flattening, respectively. The dynamic weight vector is $\mathbf{w} = \mathbf{y} \odot (w_1 - w_0) + w_0$ (with $\sigma = 10^{-12}$ as a smoothing term to avoid division by zero), where $\odot$ denotes element-wise multiplication. Using the weighted loss calculation method L_{Total} mentioned above, L_{c} and L_{f} are obtained, respectively, as follows:

$$\mathcal{L}_{\text{sum}} = \rho_1\cdot\mathcal{L}_{\text{c}} + \rho_2\cdot\mathcal{L}_{\text{f}} \tag{31}$$

where L_{sum} represents the total loss integrating coarse and fine labels, with ρ_1 and ρ_2 are custom weight parameters (ρ_1 and ρ_2 are, respectively, set to 0.2 and 1 by default).

V. Experiments and Analysis

A. Experimental Setting

To eliminate model-independent confounding factors and ensure reproducibility, stability, and consistency, all experiments in this study were conducted under a consistent server configuration. The server ran Ubuntu 20.04.6 LTS, featuring an Intel Xeon Gold 6226R CPU (2.90 GHz) as the computing core, and was equipped with 26 GB of memory to facilitate efficient data and model operations. An NVIDIA Tesla A40 GPU (48 GB) was used for acceleration, with computations and model development implemented based on the CUDA V11.6 framework and the PyTorch 1.13.0 deep learning framework.

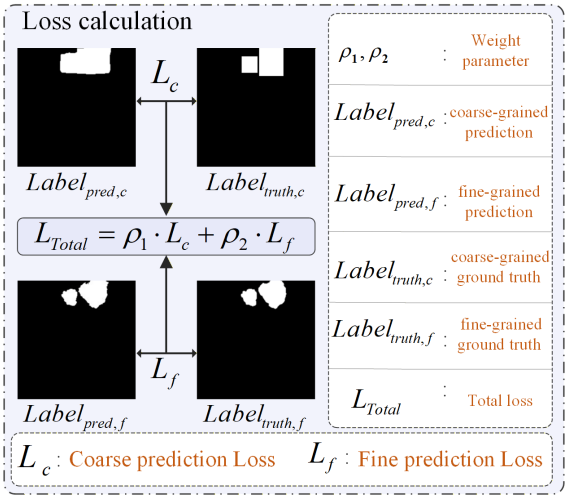


Fig. 5. L_c represents the loss calculation of the coarse-grained predictions and the coarse-grained ground truth. L_f represents the loss calculation of the fine-grained prediction and the fine-grained ground truth. L_{total} represents the total loss.

During training, random data augmentation was applied to increase sample diversity and enhance model generalization. The Adam optimizer was used to reduce gradient oscillations and promote stable convergence, while a cosine annealing strategy was employed to periodically adjust the learning rate. The learning rate started from 0 and gradually increased to a maximum initial value of $2e^{-5}$, with a batch size of 8. To prevent overfitting and conserve computational resources, training was terminated via early stopping if model performance showed no improvement over 40 consecutive epochs.

B. Evaluation Metrics

For evaluation metrics, this work employs four commonly used indices in segmentation research: Dice, mIoU, accuracy (ACC), and recall. These metrics systematically assess the semantic understanding and spatial localization capabilities of the visual–language model (VLM) in segmentation tasks from dimensions such as overlap degree, category balance, and detail sensitivity, enabling a comprehensive performance evaluation.

C. Performance Comparison and Visualization

To validate the effectiveness of the CF-SCSNet model, a series of comparative experiments was conducted on the self-built PWD dataset. The experiments compared CF-SCSNet with the state-of-the-art (SOTA) semantic segmentation models in terms of performance, number of network parameters, and computational cost. We compared the VLMs, including LViT [33], LVF [34], LAVT [20], YOLOE [50], and FIANet [41]. To verify the necessity and effectiveness of multimodality, we also compared our network model with the remote sensing image segmentation model, Clusterformer [5], UrbanSSF [4], MFNet [49] and general semantic segmentation

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TABLE I

COMPARISON OF CF-SCSNET WITH SOTA METHODS ON THE SELF-BUILT PWD DATASET, WHERE THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE UNDERLINES, RESPECTIVELY

DataType	Method	Reference	Dice(%)	mIoU(%)	Acc	Recall
V	U-Net [43]	MICCAI2015	86.15	77.23	99.15	88.50
	Segnet [44]	TPAMI2016	85.57	76.19	99.08	86.95
	deeplabv3 [45]	ECCV2018	82.79	72.42	99.04	84.53
	FCN [46]	CVPR2015	86.30	77.07	99.15	89.99
	Clusterformer [5]	TGRS2024	**88.31**	**80.23**	99.28	90.15
	UrbanSSF [4]	ISPRS2025	88.23	80.06	**99.36**	89.49
	MFNet [49]	TGRS2025	83.88	73.63	98.95	89.17
T+V	LViT [33]	TMI2023	86.52	77.24	99.01	91.91
	LVF [34]	CEA2024	85.24	75.37	98.95	91.45
	LAVT [20]	CVPR2022	85.33	75.73	99.00	90.59
	FIANet [41]	arXiv2024	86.13	76.77	99.04	91.63
	YOLOE [50]	arXiv2025	85.64	76.09	99.21	**92.08**
	CF-SCSNet	-	**90.38**	**83.55**	**99.42**	**93.64**
	Improvement		+2.07	+3.22	+0.06	+1.73

methods, U-Net [43], SegNet [44], DeepLabV3 [45], and FCN [46] by controlling variables. This comprehensive comparison aims to validate the effectiveness and superiority of CF-SCSNet across diverse model categories. The experimental results are shown in Table I. The proposed CF-SCSNet model in this article has achieved significant improvements in segmentation performance on the dataset (Dice, mIoU, and recall). Compared to the second-best model, our CF-SCSNet achieved improvements of 2.07%, 3.22%, and 1.73% in Dice, mIoU, and recall, respectively. The Dice metric is used to measure the overlap between the predicted diseased area and the real area, the mIoU metric is used to measure the mean intersection over union of disease categories, and recall is used to measure the proportion of the real diseased area that is correctly identified. Together, these three metrics reflect the precise identification and complete capture ability of the diseased area, aligning with the core goal of our focus on visual–language cross-modal fusion. Thus, the performance on these metrics demonstrates the real superiority of the method in complex scenarios. The Acc metric focuses on the accuracy of full-pixel classification. Due to the nature of the metric, existing methods have reached a saturation level of around 99%, and our model has not been optimized specifically for "full-pixel classification," so it maintains the same level as the baseline, which is a reasonable design tradeoff based on task requirements.

The comparative advantages of CF-SCSNet in segmenting three typical complex field scenarios are clearly demonstrated in the visual comparison provided in Fig. 6. First, for Sample 1, where the main challenge lies in segmenting small-scale disease lesions, models such as U-Net, MFNet, UrbanSSF, and YOLOE exhibit significant missed detections for the tiny disease spots marked by the red boxes. In contrast, the CF-SCSNet model addresses this issue through its design of differentiated receptive fields, which enables it to capture the full spectrum of lesion features, ranging from small yellow spots to large disease patches. Consequently, the segmentation results of CF-SCSNet show high consistency with the ground truth and accurately preserve the contours of the small disease spots. For Sample 2, which contains interfering elements such as other diseases or similarly colored trees, all compared models, including CF-SCSNet, exhibit some degree of mis-segmentation. However, models like U-Net, MFNet, UrbanSSF, and YOLOE contain more mis-segmented pixels within the red box and even misclassify large areas of vegetation as diseased. By employing the CFL loss strategy, CF-SCSNet is supervised to focus on both the lesion and its local spatial context, thereby avoiding distraction from irrelevant features. Consequently, when processing regions with similar colors, CF-SCSNet identifies a pixel as diseased only if it satisfies both the target's spatial distribution and its texture characteristics. This mechanism results in a significantly lower number of mis-segmented pixels compared to models that rely solely on single-feature analysis or global supervision. For Sample 3, which involves challenges of changing light angles and healthy vegetation obstruction, the YOLOE model was adapted from target detection for segmentation tasks. Its detection head prioritizes the integrity of the overall target contour, a characteristic that makes it suitable for obstructed scenarios. However, this advantage is confined to capturing the global outline, and the model exhibits significant shortcomings in pixel-level accuracy, as seen in Samples 1 and 2. Meanwhile, the other comparison models, affected by both lighting variation and obstruction, all exhibit problems such as disease omission and blurred contours. In contrast, CF-SCSNet achieves results with only minor omissions in the central-right obstructed area and successfully segments the tiny disease targets in the upper-right corner, thereby accurately covering the majority of diseased regions.

To visually demonstrate the effectiveness and stability of the proposed method, this study employs box plots to illustrate the distribution and dispersion of the experimental data, as shown in Fig. 7. In this figure, blue boxes and data points correspond to the Dice coefficient, while red ones represent the mIoU metric. Regarding the Dice, the median value of CF-SCSNet reaches 0.9038. This result is not only significantly higher than that of other compared models, but the model also exhibits a more compact box plot with a smaller interquartile range. Consequently, in terms of segmentation overlap performance, CF-SCSNet achieves the best accuracy alongside lower variability and stronger stability. Furthermore, performance in the mIoU metric reinforces this finding. In comparison,

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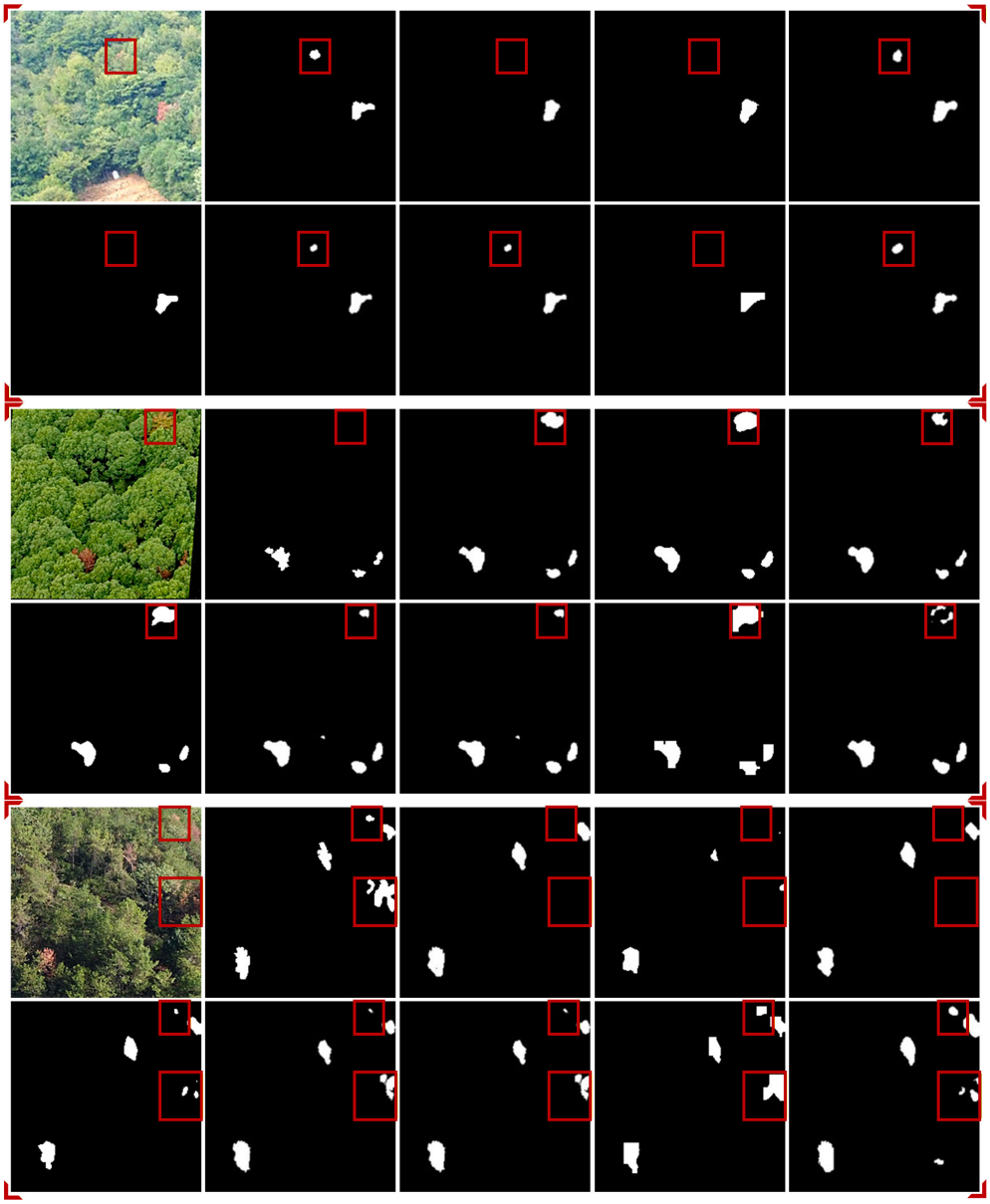


Fig. 6. Visual comparison of segmentation results on the self-built PWD dataset, contrasting the proposed CF-SCSNet with SOTA models. From left to right: RGB image, ground truth, U-Net, MFNet, Clusterformer, UrbanSSF, LViT, FIANet, YOLOE, and CF-SCSNet.

models such as MFNet and U-Net show more scattered data distributions with many low-score outliers. By contrast, CF-SCSNet displays a shorter box length and noticeably lower dispersion in its data points.

The experimental results above demonstrate the outstanding performance and robustness of our CF-SCSNet in segmenting PWD. Although some existing models, such as Clusterformer, attempt to address such complex scenarios and improve segmentation accuracy, they focus primarily on extracting disease-specific features and often fail to capture the spatial relationships between lesions and their surrounding context. As a result, they tend to exhibit high missed-detection rates for small or early-stage lesions, along with considerable pixel-level mis-segmentation.

In contrast, CF-SCSNet effectively reduces both missed detections and pixel-wise errors through the combined action of its CFL loss strategy and task-specific modules, while integrating spatial feature decoupling with multiscale detail extraction to enhance segmentation accuracy while minimizing error rates.

D. Ablation Study

To verify the efficacy of integrating the CFL and the task-customized modules (PACA and HighSF), alongside assessing the adaptability of key training setups and cross-modal components, we performed comprehensive ablation studies on the self-built PWD dataset.

1) Ablation Study on Components: We constructed a collaborative framework integrating CFL, PACA, and HighSF. Its core value lies not in the performance gain of any single component, but in how the collaboration of the three components collectively addresses the aforementioned core technical challenges. To validate this design, the proposed CF-SCSNet

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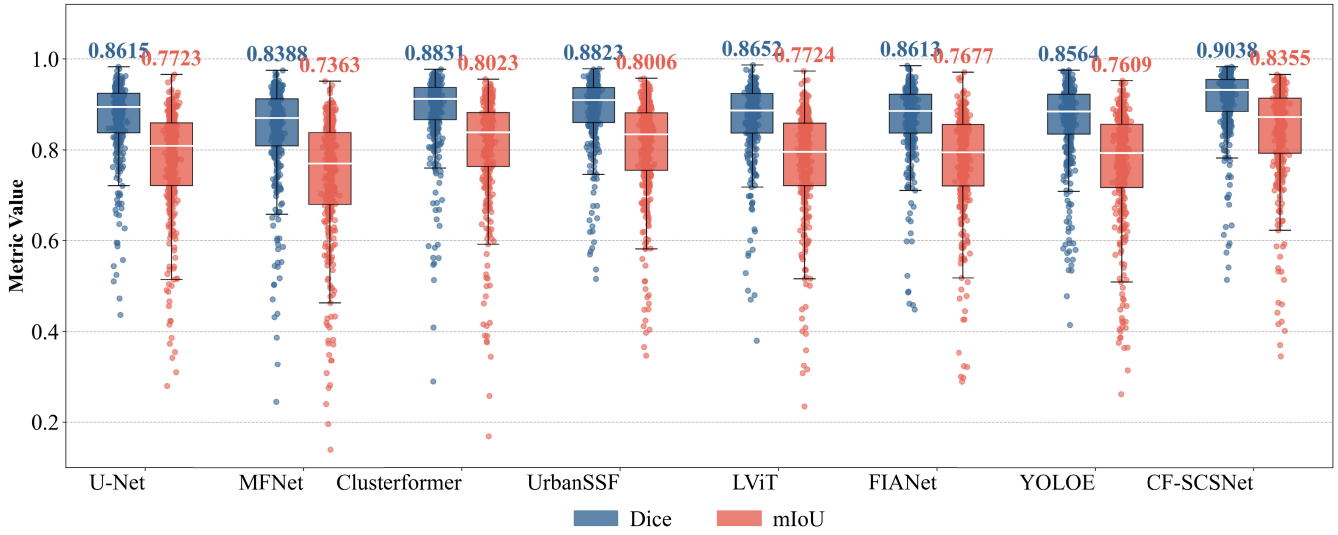


Fig. 7. Comparisons of box plots showing the values of dice and mIoU among various methods.

TABLE II
ABLATION RESULTS OF THE CF-SCSNET MODEL ON THE SELF-BUILT PWD AND THE FRAMSEG GENERALIZATION DATASETS

Method	PACA	HightSF	CFL	PWD		FramSeg	
				Dice(%)	mIoU(%)	Dice(%)	mIoU(%)
Baseline				89.50	82.18	77.05	65.95
i	✓			89.66	82.25	77.82	66.66
ii		✓		89.69	82.36	77.25	66.26
iii			✓	89.54	82.14	78.04	67.37
iv	✓	✓		90.12	83.27	77.90	66.93
v	✓		✓	89.58	82.30	78.29	67.38
vi		✓	✓	89.86	82.72	78.13	67.26
vii	✓	✓	✓	90.38	83.55	78.76	68.19

was compared against a plain baseline (without CFL, PACA, and HighSF). The focus of this comparison was on assessing the complementary roles and collaborative effect between CFL and the task-customized modules (PACA and HighSF).

From the ablation results in Table II, it can be seen that collaborative adaptation is the key to performance improvement, while the introduction of a single component fails to function effectively.

On the one hand, using CFL alone (Method iii) leads to a decrease in the model's mIoU. This is primarily because the coarse labels encompass both diseased regions and their surrounding environmental context. CFL itself, however, cannot filter and concentrate on disease-specific features, and cannot distinguish core lesions from nonlesion features. When only PACA (Method i) or HighSF (Method ii) is introduced, the model achieves only marginal performance gains. The underlying issue is that each independently operating module lacks the guidance provided by anchor points and feature selection, thereby preventing them from realizing their full potential.

On the other hand, the collaboration among modules can partially make up for the shortcomings of a single module, but there are still obvious limitations. When CFL works in conjunction with HighSF (Method vi), the nondisease environmental information carried by the coarse labels would interfere with HighSF's focus on the lesion texture. When CFL works in conjunction with PACA (Method v), PACA can collaborate with the coarse label to accurately capture the spatial correlation between the disease and the surrounding environment, while blocking the interference of nondisease environmental information and avoiding the model's excessive focus on irrelevant features. However, without the multiscale capturing ability of HighSF, it is impossible to balance the feature extraction of both large and small lesions. When the three work together (Method vii), they can effectively overcome the above limitations and significantly improve the model's positioning accuracy.

To verify the generalizability of the core component collaboration mechanism, we additionally selected the FramSeg dataset to conduct generalization ablation experiments. This dataset has a bimodal data structure combining visual and textual information, covering various terrain farmland scenarios such as plains and hills, and presenting challenges such as blurred field boundaries and interference from small features like ditches and paths. These challenges are analogous to the core difficulties in segmenting PWD. The ablation experiments in this study follow the ablation configuration of the PWD dataset, using mIoU and Dice coefficient as evaluation indicators to verify the adaptability and contribution of each component to cross-domain remote sensing segmentation tasks.

The generalized ablation results are shown in Table II. The core components continue to demonstrate significant performance gains in the agricultural field segmentation scenario, with an improvement of 4% in mIoU compared to the baseline model. This generalization ablation experiment further proves that the design of the core components of CF-SCSNet has universality, laying the groundwork for potential application in broader remote sensing tasks.

2) Analysis of the Decomposition Strategy for the PACA Module: To thoroughly validate the rationality of the decom-

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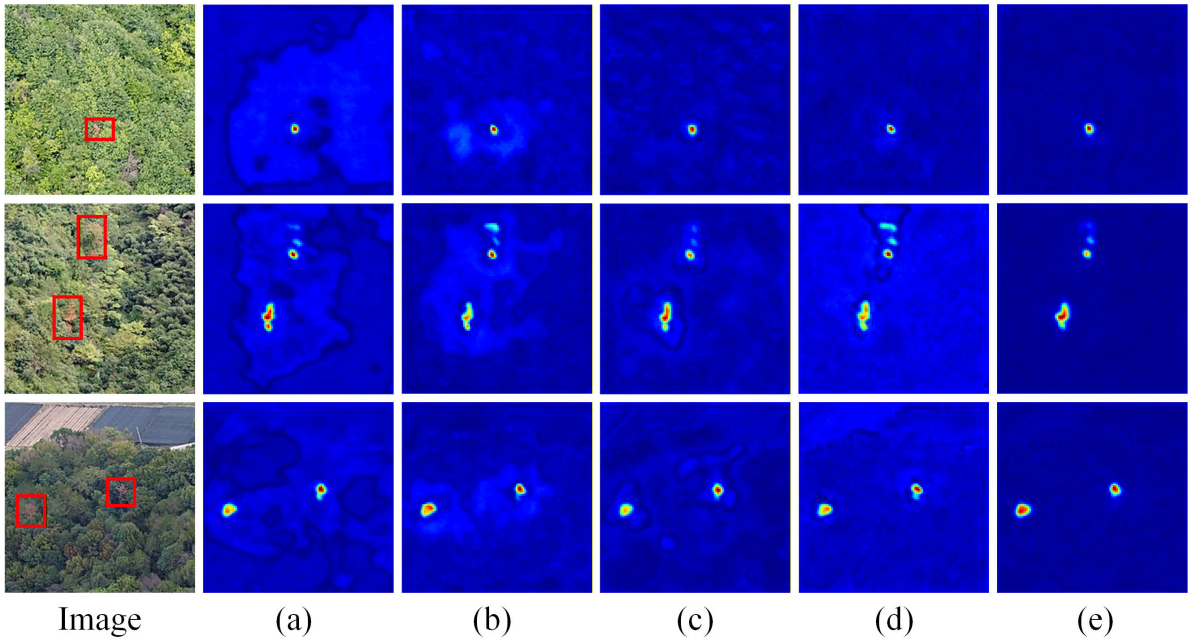


Fig. 8. Model attention heatmap (highlighted areas) for suspected diseased trees versus their actual positions (red boxes) in the original image. (a) Vertical-only group (with both average and max-pooling). (b) Horizontal-only group (with both average and max-pooling). (c) Max-only group (with both horizontal and vertical decomposition). (d) Average-only group (with both horizontal and vertical decomposition). (e) Undecomposed baseline group.

position strategy, we used the undecomposed PACA strategy as the baseline and established the following five key ablation groups for comparison. The corresponding attention heatmaps are shown in Fig. 8(a)–(e). In these visualizations, the highlighted areas represent the model's focus on suspected diseased tree regions. Our analysis reveals that unidirectional decomposition [Fig. 8(a) and (b)] tends to lose certain spatial semantic associations, while a single pooling method [Fig. 8(c) and (d)] often compromises global or local key disease signals, leading to mild highlighting of irrelevant areas around the diseased tree and thus attention dispersion. In contrast, Fig. 8(e) corresponds to the nondecomposed PACA strategy, which fully captures both the vertical lesion gradient and the horizontal distribution boundary of the diseased tree, thereby achieving precise attentional convergence to the actual diseased area and confirming the strategy's effectiveness.

3) Sensitivity Analysis of Learning Rate (LR) and Initial Restart Period (T_0): To eliminate the random interference of training configurations and ensure the reliability of model performance, we conducted ablation experiments on key hyperparameters to verify the impact of parameter values on the convergence speed, stability, and segmentation performance of the model.

We focused the ablation experiments on the core hyperparameters of the Adam optimizer combined with the CosineAnnealingWarmRestarts learning rate scheduler. Given the complex background of UAV remote sensing images, significant variations in target scales, and scattered distribution of sample features, we selected the Adam optimizer due to its strengths in adaptive learning rate adjustment and gradient oscillation suppression, which have made it a mainstream choice in this field. Its integration with the CosineAnnealingWarmRestarts scheduler can effectively meet the training

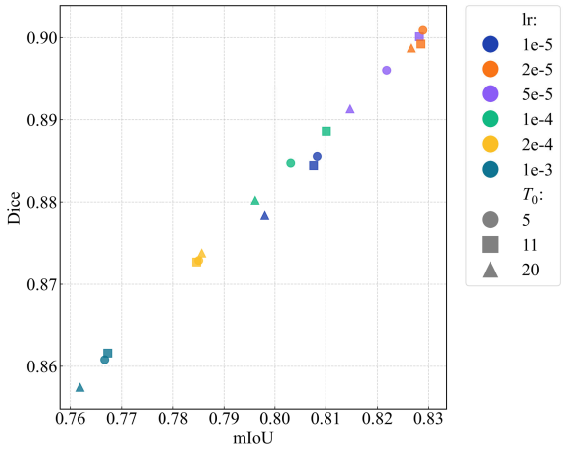


Fig. 9. 2-D distribution scatter plot of the Dice coefficient and mIoU under different lr-T_0 combinations.

demands of remote sensing models. Considering the difficulty of extracting lesions in complex field scenarios, the parameter ranges were systematically screened and optimized to exclude extreme values that could cause training failure. The initial learning rate was selected from the discrete set $1e^{-5}$, $2e^{-5}$, $5e^{-5}$, $1e^{-4}$, and $1e^{-3}$, while the initial restart cycle, T_0, was set to values from 5, 11, 20, covering the typical spectrum from short to long cycles.

To systematically evaluate the interactive effects of the learning rate (lr) and the restart cycle (T_0), we constructed a full-factorial experimental design comprising five learning rates ($1e^{-5}$, $2e^{-5}$, $5e^{-5}$, $1e^{-4}$, and $1e^{-3}$) and three cycle parameters (5, 11, and 20), resulting in 15 distinct experimental configurations.

Based on a comprehensive evaluation of the core segmentation metrics (Dice coefficient and mIoU), the combination

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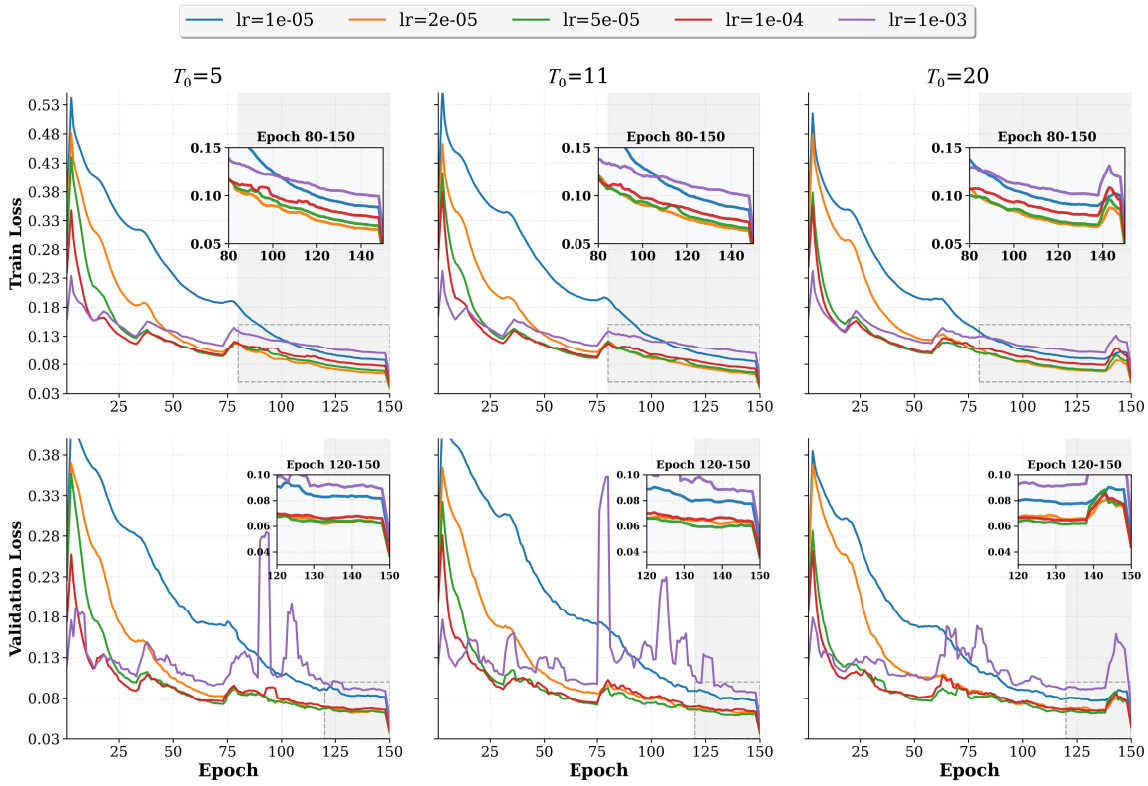


Fig. 10. Convergence curves of training/validation losses for the CF-SCSNet model under different learning rates (lr).

with lr = $2e^{-5}$ and T_0 = 5 achieved the best performance. It delivered the highest Dice (0.9009) and mIoU (0.8289) scores among all 15 experimental configurations. Moreover, the low standard deviations of these two metrics (0.0007 and 0.0012, respectively) indicate that the model attains both optimal performance and high stability. These results are directly visualized in Fig. 9, where the point corresponding to this combination lies in the optimal top-right region of the Dice–mIoU scatter plot, confirming its leading performance.

As shown in Fig. 10, a small learning rate ($1e^{-5}$) resulted in slow convergence of both training and validation loss, failing to adequately drive the model to capture the subtle texture features of PWD lesions on the limited-scale dataset. In contrast, a large learning rate ($1e^{-3}$) accelerated the training loss convergence significantly but caused large fluctuations in validation loss. This is primarily because the inherent complexity of the PWD dataset (e.g., variations in lighting and viewpoint) caused the model to overfit to local features and settle into suboptimal solutions on such a complex and heterogeneous dataset. Compared to these two extremes, learning rates of $5e^{-5}$ and $1e^{-4}$ improved convergence speed without severe oscillation, indicating a better match with the dataset's basic characteristics. However, both still exhibited minor fluctuations during training and validation, and their stability was lower than that of lr = $2e^{-5}$, which demonstrated a steady and consistent decline in both training and validation loss.

4) Performance Comparison of Different Pretrained Text Encoders: To fully demonstrate the rationale for selecting RoBERTa-base as the text encoder, we compared its performance with three other widely used pretrained language models in vision–language tasks: BERT-base, DistilBERT-base, and XLNet-base. DistilBERT-base serves as a lightweight encoder designed to improve inference speed and reduce resource consumption through an optimized architecture. XLNet-base, in contrast, offers distinct advantages in long-text semantic modeling. Throughout the experiments, variables were strictly controlled. Each encoder was evaluated over three independent experimental runs, and the mean and variance of the results were calculated.

TABLE III

PERFORMANCE COMPARISON OF DIFFERENT PRETRAINED TEXT ENCODERS. THE UNDERLINED AND BOLDED PARTS INDICATE THE BEST OPTION

Text Encoder	Dice mean(%)↑	Dice std↓	mIoU mean(%)↑	mIoU std↓
BERT-base	90.05	0.05	**83.01**	0.13
RoBERTa-base	**90.13**	0.06	82.96	0.09
DistilBERT-base	89.81	**0.02**	82.90	**0.04**
XLNet-base	89.90	0.09	82.91	0.14

The experimental results are shown in Table III. In terms of the core segmentation metrics, the mean Dice of RoBERTa-base reaches 90.13%, which is the highest among all compared models. It exceeds DistilBERT-base by 0.32% and XLNet-base by 0.23%. This result directly reflects the optimal performance of RoBERTa in terms of overlap accuracy for target region segmentation. Regarding the mIoU metric, the mean value of RoBERTa-base is 82.96%. Although slightly lower than the 83.01% of BERT-base, it remains at a relatively high level. Its

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standard deviation for mIoU is 0.09%, which is higher than that of BERT-base, demonstrating good stability.

Further analysis indicates that the text data in this study exhibit significant variation in length, a characteristic that imposes a stringent requirement on the encoder's semantic robustness. RoBERTa-base, employing a dynamic masking strategy and benefiting from large-scale pretraining, stably captures fine-grained semantic features when processing texts of mixed lengths. In contrast, although BERT-base achieves a slightly higher mean mIoU, its static masking strategy is prone to overfitting at masked positions, thereby restricting its semantic modeling capability across texts of varying lengths. While DistilBERT-base shows the smallest variance and optimal stability, its means for both Dice and mIoU are comparatively low. XLNet-base's strength in long-text modeling relies on complete, untruncated text input. Under the unified encoding configuration adopted in this experiment, where original long texts are truncated, and short texts are padded, its long-text modeling capability fails to be effectively utilized.

E. Generalization Experiments

1) FarmSeg and RRSIS Datasets: To advance the field segmentation of PWD via UAV remote sensing, the present study focuses on investigating the cross-scene generalization capability of segmentation models.

Given the limited diversity of background scenarios in dedicated PWD datasets, two heterogeneous datasets are introduced for cross-scene generalization validation. Specifically, the FramSeg dataset captures the scenario of small, irregular subregions distributed within large, homogeneous areas. Accordingly, it serves to verify the model's capability in locating small targets within visually consistent yet complex environments, thereby aligning to detect scattered infected pines in forest stands. Conversely, the urban building dataset requires segmenting small-scale structures, such as low-rise buildings and auxiliary facilities, from cluttered backgrounds with multiple textures and scales. Hence, this dataset effectively evaluates the model's robustness in suppressing multiscale noise, simulating the interference posed by analogous objects in forest scenarios.

Owing to the large scale of the public FarmSeg [47] dataset, which contains 15 638 training and 4519 validation images (512×512 pixels, covering eight geographical regions), the verification of model generalization becomes computationally inefficient. To improve efficiency, this study employed a stratified sampling method. A representative subset was constructed by randomly selecting 2000 training, 300 validation, and 562 test images from each region, ensuring the preservation of both geographical diversity and scenario specificity across all eight regions. Similarly, considering the computational burden posed by the high-resolution images in the RRSIS [48] dataset (17 402 images, 800×800 pixels), a stratified random sampling strategy based on building categories was adopted. This strategy yielded a representative subset of 1600 images, comprising 1000 for training, 200 for validation, and 400 for testing. Consequently, the sample category distribution remains consistent with the original dataset, achieving a balance between verification efficiency and the reliability of the generalization evaluation.

TABLE IV

PERFORMANCE COMPARISON OF CF-SCSNET WITH OTHER METHODS ON FRAMSEG AND RRSIS PUBLIC DATASETS. "M" DENOTES SMALL TARGET DATASET, AND "L" DENOTES THE LARGE TARGET DATASET. THE BEST RESULT IS IN RED FONT WITH AN UNDERLINE

Method	FramSeg-M / RRSIS-M		FramSeg-L / RRSIS-L	
	Dice	mIoU	Dice	mIoU
U-Net [43]	72.69 / 37.08	61.13 / 28.70	85.92 / 41.87	78.26 / 31.99
Segnet [44]	72.89 / 39.89	61.15 / 31.32	83.44 / 44.64	74.97 / 34.55
deeplabv3 [45]	73.00 / 39.60	61.24 / 30.96	84.93 / 44.58	76.92 / 34.22
LViT [33]	77.29 / 45.74	66.02 / 35.64	86.78 / 53.68	79.44 / 42.77
LAVT [20]	72.29 / 45.71	59.99 / 36.01	83.76 / 52.59	75.35 / 42.07
CF-SCSNet	**78.76 / 50.04**	**68.19 / 40.46**	**88.15 / 57.73**	**81.36 / 47.70**

2) Small-Target Dataset Construction Strategy: To specifically evaluate the model's capability in segmenting small targets and to address the common problem of missed detections in existing remote sensing segmentation models, a further screening of the FarmSeg and RRSIS datasets was conducted. Accordingly, the training, validation, and test sets were filtered based on a criterion where the proportion of foreground pixels in the mask file was less than 5%. This filtering step was applied in line with the aforementioned stratified sampling strategy.

The visual results in Fig. 11 and the experimental results in Table IV are mutually consistent, collectively demonstrating the performance advantages of CF-SCSNet over the comparison models. In terms of overall performance, these comparison models commonly exhibit issues such as missed detections or pixel-level mis-segmentation in small-target scenarios, and frequently produce blurred, fragmented, or over-segmented outputs in scenarios with complex boundaries. Some VLMs also show performance degradation in cross-geographical scenarios. In contrast, CF-SCSNet can reliably identify low-proportion targets (Samples 1 and 5), accurately restore target contours and texture details, and maintain robust segmentation accuracy across diverse geographical settings (Samples 2, 3, and 6).

VI. DISCUSSION

In remote sensing segmentation of PWD, early-stage and small-sized lesions exhibit weak features that are easily obscured by the background, and their spatial correlation with the environment is also difficult to capture. Existing methods often overlook subtle features and spatial context, leading to high missed-detection rates. To address this, we propose CF-SCSNet, which integrates a collaborative CFL supervision design with customized PACA and HighSF modules to leverage textual semantic guidance for fine-grained segmentation.

We evaluate CF-SCSNet on UAV-based PWD segmentation across three scenarios: 1) specialized segmentation (compared with Clusterformer); 2) lightweight edge deployment (compared with UrbanSSF and YOLOE); and 3) vision–language understanding (compared with LViT). Key performance indicators include parameter count (Params),

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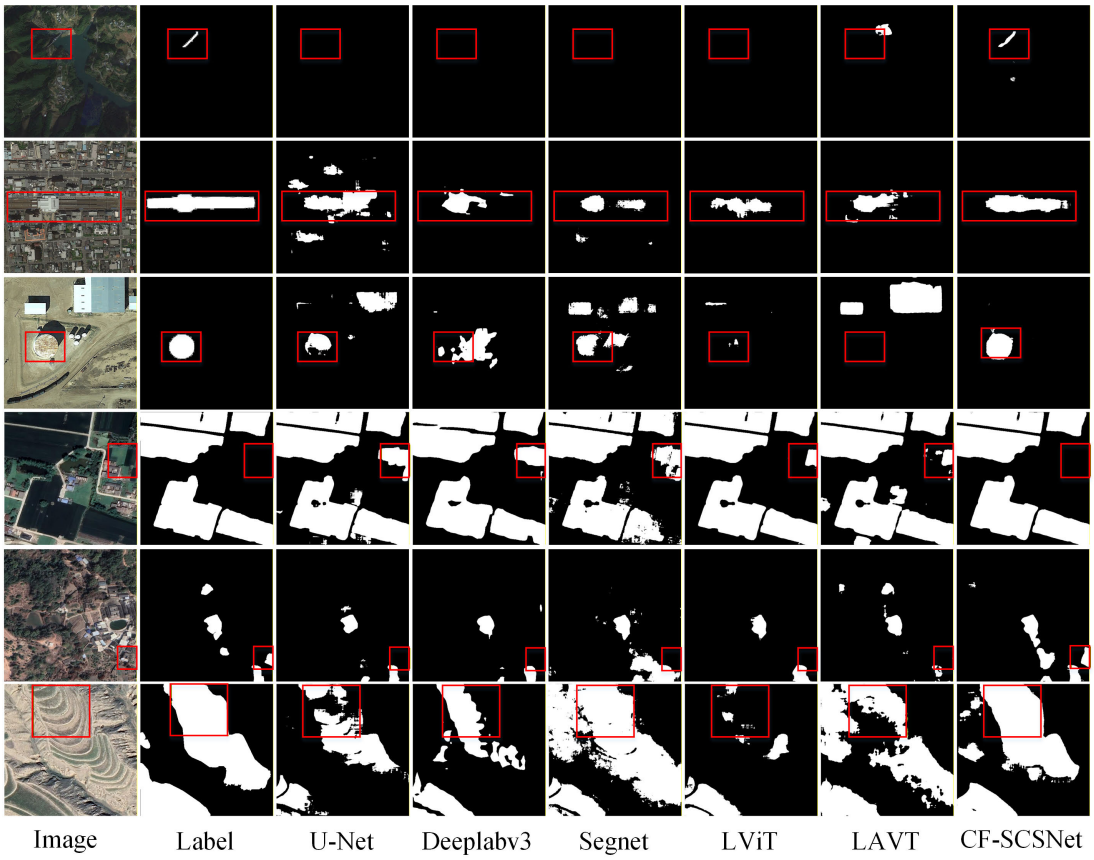


Fig. 11. Visual comparison of different remote sensing image segmentation models on the RRSIS and FramSeg datasets. "label" denotes the ground truth.

TABLE V

PERFORMANCE AND COMPUTATIONAL EFFICIENCY COMPARISON OF DIFFERENT MODELS ON THE PWD DATASET. THE UNDERLINED AND BOLDED PARTS INDICATE THE BEST OPTION

Method	Memory (GB)↓	FPS↑	FLOPs (G)↓	Parm (M)↓	mIoU (%)↑
Clusterformer	5.04	24.87	6.77	11.34	80.23
UrbanSSF	2.39	**72.11**	**1.80**	**3.48**	80.06
LViT	6.34	44.64	52.52	25.05	77.24
YOLOE	**1.83**	51.77	12.91	45.94	76.09
CF-SCSNet	5.31	29.41	47.98	35.26	**83.55**

floating-point operations (FLOPs), memory usage (Memory), and inference speed (FPS).

As shown in Table V, due to cross-modal attention and vision–language interaction modules, CF-SCSNet has larger FLOPs and parameters than unimodal models, leading to higher GPU memory usage and lower FPS. We regard this computational overhead as a reasonable tradeoff for performance improvement. With bimodal semantic supervision, the model improves mIoU by 3.32%, and its parallel architecture keeps the average single-image inference time at 53.35 ms, balancing speed and resource consumption.

Regarding the reliability of the proposed method, the results listed in Table V are averaged over independent experimental runs, with all variances remaining consistently below 1.5%. These consistent results demonstrate that the performance improvements are stable rather than caused by random fluctuations. Meanwhile, the stable inference time (53.35 ms) and controllable memory consumption reflect the method's efficiency, which is critical for applications such as UAV-based remote sensing processing.

Nevertheless, the high accuracy of CF-SCSNet depends on fine-grained text annotations, which are difficult to obtain quickly in real forestry inspections. Without standardized text support, model performance may degrade, accompanied by high annotation costs and practical application bottlenecks. Therefore, future work will explore an LLM-based automated prompt generation framework. Using a small number of expert annotations, we will construct a lightweight prompt template library to reduce reliance on manual, detailed text labeling and ease the difficulty of text acquisition in field applications.

VII. CONCLUSION

To address the core challenges in PWD segmentation posed by subtle early lesion features and occlusion of small targets in complex backgrounds, we built a UAV remote sensing dataset for PWD in complex field scenarios and proposed the CF-SCSNet model. By integrating coarse and fine label supervision with task-specific modules, the model confirms that textual semantic guidance effectively improves segmentation accuracy. Experimental results demonstrate that our method surpasses multiple advanced visual models in segmentation performance, obtaining an mIoU of 83.55% on the self-built PWD dataset. Since this study focuses on verifying the value of textual semantic guidance rather than pursuing lightweight deployment, the model achieves superior performance but suffers from certain limitations: structural complexity, high

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computation costs, and reliance on fine-grained text annotations. In future work, we will further optimize the cross-modal fusion mechanism and simplify the model structure toward more efficient and practical UAV-based PWD detection.

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